VERILOG -מעבדה 2

מגישים

אלעד אסבג&יוסי ברים

ATP

נבדוק את זמני הכניסות כדי לקבל את כל האפשרויות.

נתחיל מהשעון . זמן מחזור של 10 ננו שניות לכן הוא התהפך כל 5 ננו שניות.

את הריסט נגדיר כ1 למשך 100 ננו שניות ולאחר מכן 0 כדי לקבל סינכרון.

ואת הסל נתחיל ב0 למשך 260 ננו שניות כדי לראות את 2 המצבים וכדי שלא יהיה בדיוק זמן מחזור שלם

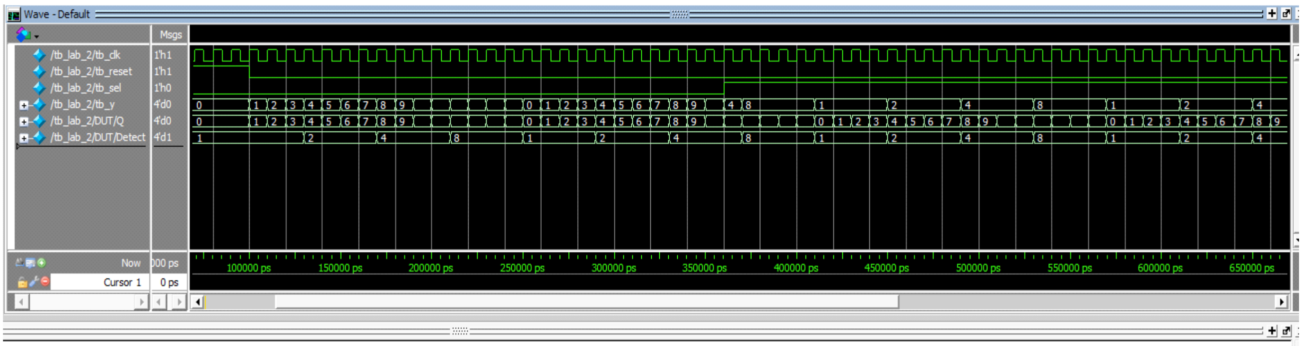
ונסתכל על התוצאות בסימולציה.

התוכנית:

```
module lab_2
(
 input          clk   ,
 input          reset ,
 input          sel   ,
 output         [3:0] y
);
reg [3:0] Q ;
wire [3:0] Detect;

always @ (posedge clk)
    if (reset)
        Q <= 4'b0;
    else
        Q <= Q + 1;


// DECODER
assign Detect[0] = (Q >= 4'd0 ) && (Q <= 4'd3 )   ? 1'b1 : 1'b0;
assign Detect[1] = (Q >= 4'd4 ) && (Q <= 4'd7 )   ? 1'b1 : 1'b0;
assign Detect[2] = (Q >= 4'd8 ) && (Q <= 4'd11)   ? 1'b1 : 1'b0;
assign Detect[3] = (Q >= 4'd12) && (Q <= 4'd15)   ? 1'b1 : 1'b0;

/// MUX
assign y = (sel) ? Detect : Q;

endmodule

```

TestBench:

```
`timescale 1ns/1ps

module tb_lab_2
(
);

reg  tb_clk, tb_reset, tb_sel;
wire     [0:3]  tb_y;

initial begin
    $display ("time\t  clk     reset    sel    q   detect     y");
    $monitor ("%8g\t\t%b        %b        %b     %h     %h          %h",
    $time, tb_clk, tb_reset , tb_sel, DUT.Q, DUT.Detect, tb_y);

    tb_clk          = 1;
    tb_reset        = 1;
    tb_sel          = 0;
    #100   tb_reset = 0;
    #260   tb_sel   = 1;
end

always begin

    #5 tb_clk = ~tb_clk;
end

lab_2 DUT
(
.clk        (tb_clk   ),
.reset      (tb_reset),
.sel        (tb_sel   ),
.y          (tb_y     )
);

endmodule
```

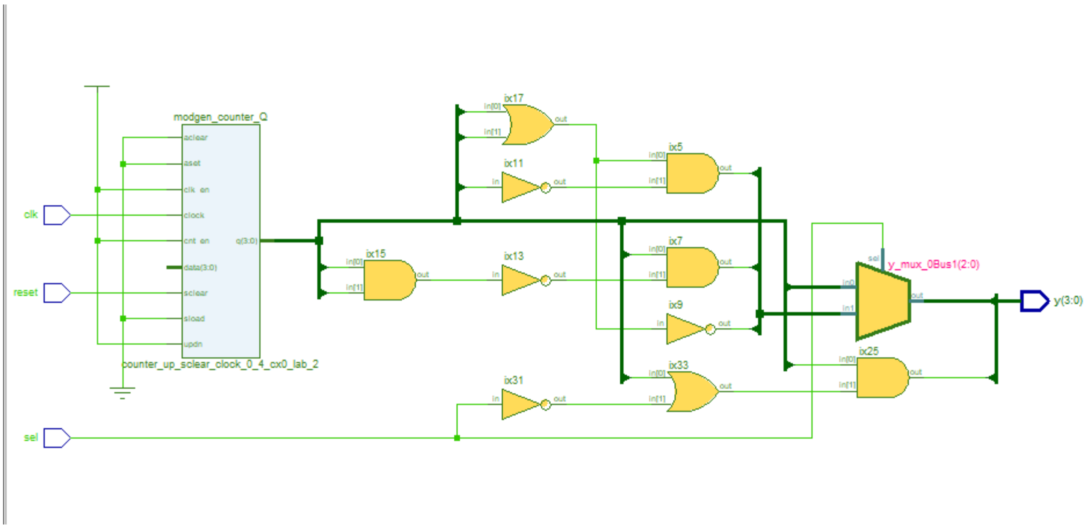


:הסימולציה

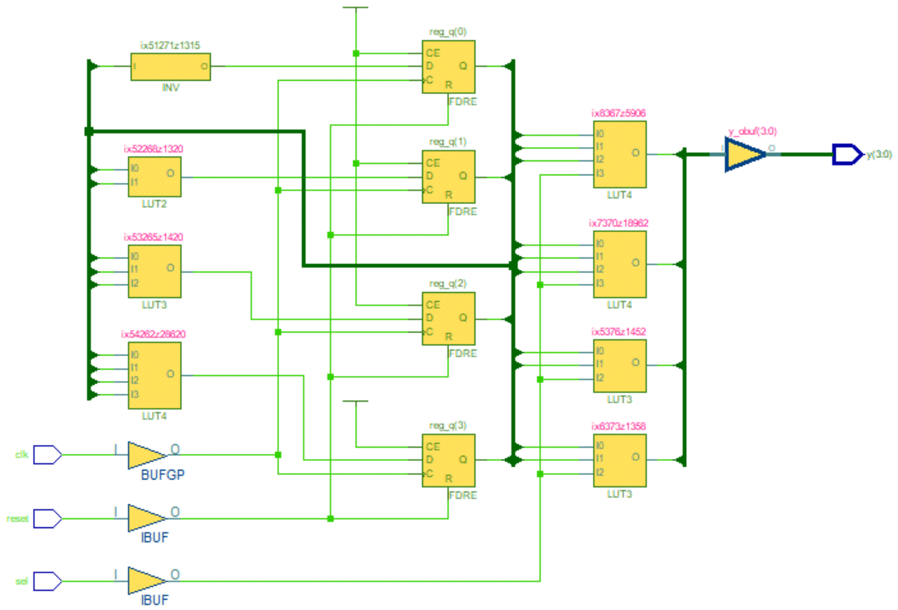
clk	reset	sel	q	detect	y	# time
1	1	0	0	1	0	# 0
0	1	0	0	1	0	# 5
1	1	0	0	1	0	# 10
0	1	0	0	1	0	# 15
1	1	0	0	1	0	# 20
0	1	0	0	1	0	# 25
1	1	0	0	1	0	# 30
0	1	0	0	1	0	# 35
1	1	0	0	1	0	# 40
0	1	0	0	1	0	# 45
1	1	0	0	1	0	# 50
0	1	0	0	1	0	# 55
1	1	0	0	1	0	# 60
0	1	0	0	1	0	# 65
1	1	0	0	1	0	# 70
0	1	0	0	1	0	# 75
1	1	0	0	1	0	# 80
0	1	0	0	1	0	# 85
1	1	0	0	1	0	# 90
0	1	0	0	1	0	# 95
1	0	0	1	1	1	# 100
0	0	0	1	1	1	# 105
1	0	0	2	1	2	# 110
0	0	0	2	1	2	# 115
1	0	0	3	1	3	# 120
0	0	0	3	1	3	# 125
1	0	0	4	2	4	# 130
0	0	0	4	2	4	# 135
1	0	0	5	2	5	# 140
0	0	0	5	2	5	# 145
1	0	0	6	2	6	# 150
0	0	0	6	2	6	# 155

1	0	0	7	2	7	#	160
0	0	0	7	2	7	#	165
1	0	0	8	4	8	#	170
0	0	0	8	4	8	#	175
1	0	0	9	4	9	#	180
0	0	0	9	4	9	#	185
1	0	0	a	4	a	#	190
0	0	0	a	4	a	#	195
1	0	0	b	4	b	#	200
0	0	0	b	4	b	#	205
1	0	0	c	8	c	#	210
0	0	0	c	8	c	#	215
1	0	0	d	8	d	#	220
0	0	0	d	8	d	#	225
1	0	0	e	8	e	#	230
0	0	0	e	8	e	#	235
1	0	0	f	8	f	#	240
0	0	0	f	8	f	#	245
1	0	0	0	1	0	#	250
0	0	0	0	1	0	#	255
1	0	0	1	1	1	#	260
0	0	0	1	1	1	#	265
1	0	0	2	1	2	#	270
0	0	0	2	1	2	#	275
1	0	0	3	1	3	#	280
0	0	0	3	1	3	#	285
1	0	0	4	2	4	#	290
0	0	0	4	2	4	#	295
1	0	0	5	2	5	#	300
0	0	0	5	2	5	#	305
1	0	0	6	2	6	#	310
0	0	0	6	2	6	#	315
1	0	0	7	2	7	#	320
0	0	0	7	2	7	#	325
1	0	0	8	4	8	#	330
0	0	0	8	4	8	#	335
1	0	0	9	4	9	#	340
0	0	0	9	4	9	#	345
1	0	0	a	4	a	#	350
0	0	0	a	4	a	#	355
1	0	1	b	4	4	#	360
0	0	1	b	4	4	#	365
1	0	1	c	8	8	#	370
0	0	1	c	8	8	#	375

1	0	1	d	8	8	#	380
0	0	1	d	8	8	#	385
1	0	1	e	8	8	#	390
0	0	1	e	8	8	#	395
1	0	1	f	8	8	#	400
0	0	1	f	8	8	#	405
1	0	1	0	1	1	#	410
0	0	1	0	1	1	#	415
1	0	1	1	1	1	#	420
0	0	1	1	1	1	#	425
1	0	1	2	1	1	#	430
0	0	1	2	1	1	#	435
1	0	1	3	1	1	#	440
0	0	1	3	1	1	#	445
1	0	1	4	2	2	#	450
0	0	1	4	2	2	#	455
1	0	1	5	2	2	#	460
0	0	1	5	2	2	#	465
1	0	1	6	2	2	#	470
0	0	1	6	2	2	#	475
1	0	1	7	2	2	#	480
0	0	1	7	2	2	#	485
1	0	1	8	4	4	#	490
0	0	1	8	4	4	#	495



RTL

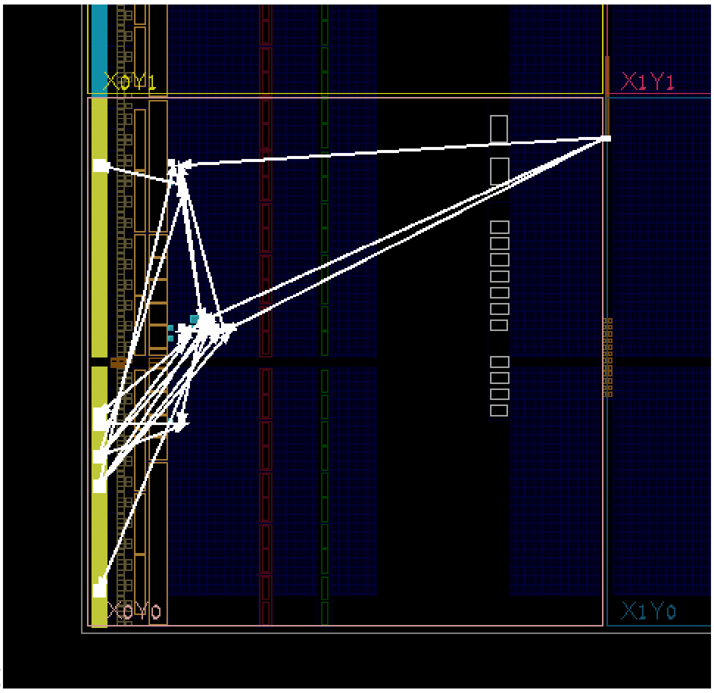


TECHNOLOGY SCHEMATICS:

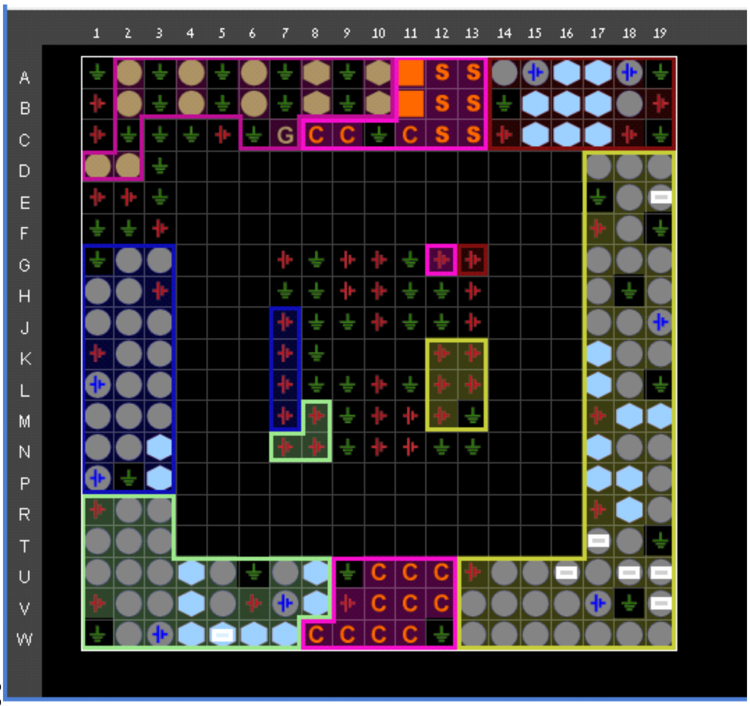
XDC:

```
#################################################################################
# Mentor Graphics Corporation
#
#################################################################################


################
# create clock #
################

# Precision Generated
create_clock [get_ports {clk}] -name {clk} -period 10 -waveform {0 5}
create_clock  -name {VirtualClock} -period 10 -waveform {0 5}

###################
# set input delay #
###################

# Precision Generated
set_input_delay 0 -clock [get_clocks {VirtualClock}] -add_delay  [get_ports {reset sel}]

####################
# set output delay #
####################

# Precision Generated
set_output_delay 0 -clock [get_clocks {VirtualClock}] -add_delay  [get_ports {y*}]

##############################
# Pin Locations and Voltages #
##############################

set_property -dict { PACKAGE_PIN W5   IOSTANDARD LVCMOS33 } [get_ports { clk   }];
set_property -dict { PACKAGE_PIN U18  IOSTANDARD LVCMOS33 } [get_ports { reset }];
set_property -dict { PACKAGE_PIN T17  IOSTANDARD LVCMOS33 } [get_ports { sel }];
set_property -dict { PACKAGE_PIN U16   IOSTANDARD LVCMOS33 } [get_ports { y*[0]  }];
set_property -dict { PACKAGE_PIN E19   IOSTANDARD LVCMOS33 } [get_ports { y*[1]  }];
set_property -dict { PACKAGE_PIN U19   IOSTANDARD LVCMOS33 } [get_ports { y*[2]  }];
set_property -dict { PACKAGE_PIN V19   IOSTANDARD LVCMOS33 } [get_ports { y*[3]  }];



##########################
# Configuration Settings #
##########################
set_property CFGBVS VCCO [current_design]
set_property CONFIG_VOLTAGE 3.3 [current_design]
```

Device View:



Pin Location:



Summary

		Hold		Pulse Width	
egative Slack (WNS):	-2.239 ns	Worst Hold Slack (WHS):	0.256 ns	Worst Pulse Width Slack (WPWS):	4.500 ns
gative Slack (TNS):	-7.747 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
of Failing Endpoints:	4	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
mber of Endpoints:	14	Total Number of Endpoints:	14	Total Number of Endpoints:	6

traints are not met.

Timing :